$2Y = X+50$

$Y = \frac{1}{2}X+25$ therefore $b_{yx} = \frac{1}{2}$

Clearly equation (2) would be treated as regression equation of X on Y

$3Y-2X-10 = 0$

$2X = 3Y-10$

$X = \frac{3}{2}Y-5$ therefore $b_{xy} = \frac{3}{2}$

The Correlation coefficient $r = \pm\sqrt{b_{xy} \times b_{yx}}$

$r = \sqrt{\frac{1}{2} \times \frac{3}{2}} = 0.866$

NOTE

It may be noted that in the above problem one of the regression coefficient is greater than 1 and the other is less than 1. Therefore our assumption on given equations are correct.

Example 9.13

Find the means of X and Y variables and the coefficient of correlation between them from the following two regression equations:

$4X-5Y+33 = 0$

$20X-9Y-107 = 0$

Solution:

We are given

$4X-5Y+33 = 0$ ... (1)

$20X-9Y-107 = 0$ ... (2)

Solving equation (1) and (2)

We get $Y = 17$

Putting the value of Y in equation (1)

We get $X = 13$

Hence $\overline{X} = 13$ and $\overline{Y} = 17$

Calculating correlation coefficient

Let us assume equation (1) be the regression equation of X on Y

$4X = 5Y-33$

$X = \frac{1}{4}(5Y-33)$

$X = \frac{5}{4}Y-\frac{33}{4}$

$b_{xy} = \frac{5}{4} = 1.25$

Let us assume equation (2) be the regression equation of Y on X

$9Y = 20X-107$

$Y = \frac{1}{9}(20X-107)$

$Y = \frac{20}{9}X-\frac{107}{9}$

$b_{yx} = \frac{20}{9} = 2.22$

But this is not possible because both the regression coefficient are greater than 1. So our above assumption is wrong. Therefore treating equation (1) has regression equation of Y on X and equation (2) has regression equation of X on Y. So we get

$b_{yx} = \frac{4}{5} = 0.8$ and

$b_{xy} = \frac{9}{20} = 0.45$

The Correlation coefficient

$r = \pm\sqrt{b_{xy} \times b_{xy}}$

$r = \sqrt{0.45 \times 0.8} = 0.6$

Example 9.14

The following table shows the sales and advertisement expenditure of a form

Title	Sales	Advertisement expenditure (₹ in Crores)
Mean	40	6
SD	10	1.5

09_11th_BM-STAT_Ch-9-EM.indd 223 21-04-2020 12:28:33 PM